

Mini-Briefing

Pattern Matching and Machine Learning for Audio Signal Processing

MA-INF 2212

#1 Cyclic Operators

Prof. Dr. Frank Kurth



Cyclic Operators: Why?

Example:

„cyclic warp-around“

$$T_m x(n) := x(n-m \bmod N)$$

- Easier way of defining & handling operators such as shifts for **finite** signals
- Compatible with certain math. tools/results, most importantly, the Convolution Theorem for finite signals
(i.e. the Convolution Theorem for finite signals only works with cyclic convolution)

1. Infinite signals (non-cyclic case)

Values	...	1	3	4	-5	2	1	...		x
Indexes	...	-1	0	1	2	3	4	5	6	...
2-shifted values			...	1	3	4	-5	2	1	...
										T_2x

Note: A red arrow labeled '2' points from index 0 to index 2. A green arrow points from the value 4 at index 3 in the second row to the value 4 at index 3 in the third row.

Definition:

$$T_m x(n) := x(n-m)$$

Example:

$$T_2 x(3) = x(3-2) = x(1) = 4$$

2. Finite signals

Values	[	1	3	4	-5	2	]	
Indexes		0	1	2	3	4	5	6
		→			→			
2-shifted values	[	?	?	1	3	4	-5	2

2. Finite signals

Values	[	1	3	4	-5	2	]	x
Indexes		0	1	2	3	4	5	6
Cyclic shift	[	-5	2	1	3	4	]	T_2x

Definition:

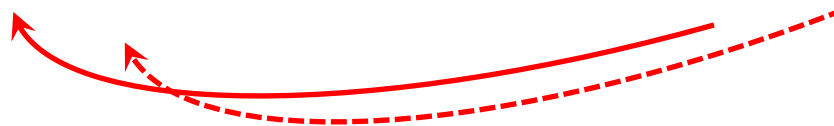
$$T_m x(n) := x(n - m \bmod N)$$

Example:

$$T_2 x(1) = x(1 - 2 \bmod 5) = x(-1 \bmod 5) = x(4) = 2$$

2. Finite signals

Values	[	1	3	4	-5	2	]			x
Indexes		0	1	2	3	4		5	6	
Cyclic shift	[	-5	2	1	3	4	]	-5	2	T_2x
Cyclic indexes		5	6	7	8	9				



2. Finite signals

$$T_m x(n) := x(n - m \bmod N)$$

	-5	-4	-3	-2	-1	...
Indexes	0	1	2	3	4	
Equivalent indexes	5	6	7	8	9	
	10	11	12	13	14	
	...					

Definition: $k \bmod N$:= unique $0 \leq a < N$ such that $a + rN = k$ for an integer r

Example: $1 = 6 \bmod 5 = 11 \bmod 5 = -4 \bmod 5$

$$ACF[x](m) = \langle x | T_m x \rangle$$

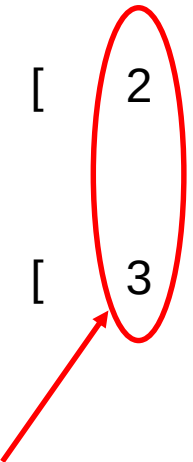
x	$=$	[	2	0	0	1	3	]		
$T_0 x$	$=$	[	2	0	0	1	3	]	0	14
$T_1 x$	$=$	[	3	2	0	0	1	]	1	9
$T_2 x$	$=$	[	1	3	2	0	0	]	2	2
$T_3 x$	$=$	[	0	1	3	2	0	]	3	2
$T_4 x$	$=$	[	0	0	1	3	2	]	4	9

But wait, is cyclicity what we really want?

Cyclic „artifacts“: Also cyclical Patterns are found/detected

$$ACF[x](m) = \langle x | T_m x \rangle$$

x	$=$	[	2	0	0	1	3	]	m	$ACF[x](m)$
$T_1 x$	$=$	[	3	2	0	0	1	]	1	9



Compares cyclically shifted „past“
(3) and „present“ (2)

If this is not desirable due to the applications, **zero padding** of x by $N = \text{length}(x)$ samples can be used ($x' = [2 \ 0 \ 0 \ 1 \ 3 \ 0 \ 0 \ 0 \ 0 \ 0]$)

- Easier way of defining & handling operators such as shifts for **finite** signals
- Compatible with certain math. tools/results, most importantly, the Convolution Theorem for finite signals
(i.e. the Convolution Theorem for finite signals only works with cyclic convolution)
- Cyclic shift: samples shifted outside the array to the „right“, are re-inserted to the „left“
- Cyclic Artifacts can be avoided using zero padding